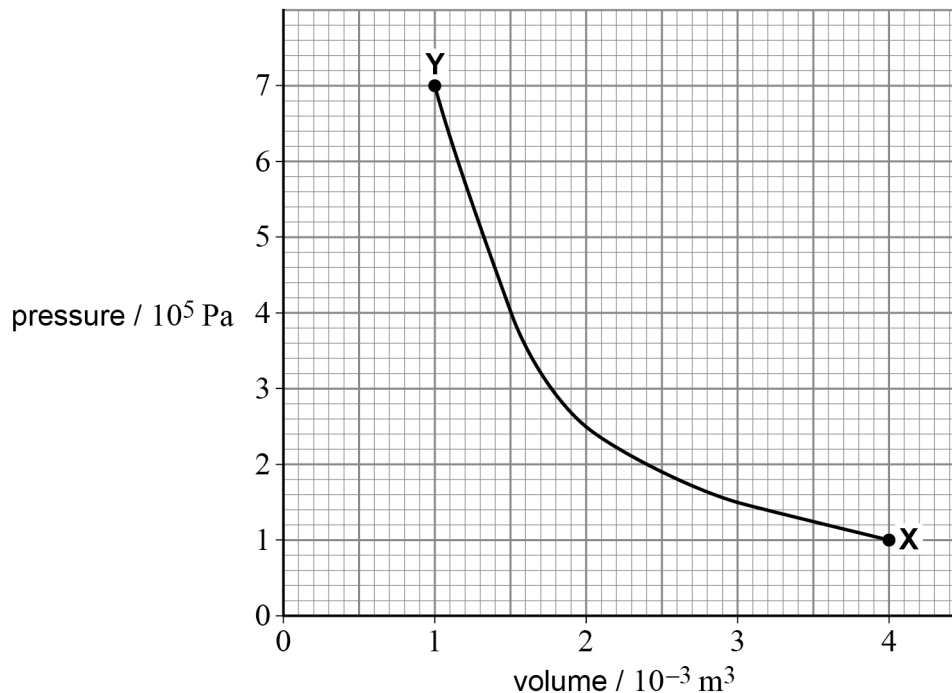


The graph shows the variation of pressure with volume for a fixed mass of gas as it changes from state **X** to state **Y**.

T_X and T_Y are the temperatures of the gas at states **X** and **Y** respectively.

U_X and U_Y are the internal energies of the gas at states **X** and **Y** respectively.



Which row describes the temperatures and internal energies at **X** and **Y**?

[1 mark]

	Temperatures	Internal energies	
A	$T_Y < T_X$	$U_Y = U_X$	☐
B	$T_Y < T_X$	$U_Y < U_X$	☐
C	$T_Y > T_X$	$U_Y = U_X$	☐
D	$T_Y > T_X$	$U_Y > U_X$	☐

Which is the equivalent of the internal energy of 1 mol of an ideal gas with a temperature T ?

[1 mark]

A $\frac{2}{3}RT$



B $\frac{3}{2}RT$



C $\frac{2}{3} \frac{RT}{N_A}$



D $\frac{3}{2} \frac{RT}{N_A}$



Turn over for the next question

Turn over ►



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Which row describes the potential and kinetic energies of particles in a solid material at a temperature of absolute zero?

[1 mark]

	Potential energy	Kinetic energy	
A	0	0	☐
B	0	< 0	☐
C	< 0	0	☐
D	< 0	< 0	☐

1 3

The table shows data for two ideal gases, **X** and **Y**.

X and **Y** each have the same mass and volume.

	X	Y
Pressure / kPa	100	200
Molecular mass	14	28
Temperature / °C	−23	

What is the temperature of gas **Y**?

[1 mark]

A 727 °C

☐

B 227 °C

☐

C −23 °C

☐

D −148 °C

☐

Don't write

1 4

A student wants to do an experiment to verify Boyle's law.

Which quantities must be measured and which quantity must be kept constant?

[1 mark]

	Quantities to be measured	Quantity to be kept constant	
A	Pressure and temperature	Volume	☐
B	Pressure and volume	Temperature	☐
C	Volume and temperature	Pressure	☐
D	Pressure, volume and temperature	None	☐

- A** the pressure of a fixed mass of gas at constant volume depends on the absolute temperature. ☐
- B** the average molecular kinetic energy of the molecules of a gas depends on the absolute temperature. ☐
- C** gases are made up of randomly moving particles which have momentum. ☐
- D** in ideal gases, the internal energy is the same as the kinetic energy of the particles. ☐



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1	6
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The average molecular kinetic energy of the molecules in a sample of gas is $\frac{1}{2}m(c_{\text{rms}}^2)$
where m is the mass of one molecule.

What is the magnitude of the average velocity of the molecules in the sample of gas?

[1 mark]

A $\sqrt{2} \times c_{\text{rms}}$

☐

B c_{rms}

☐

C $\frac{c_{\text{rms}}}{\sqrt{2}}$

☐

D 0

☐

Two containers, **X** and **Y**, contain helium gas. The rms speed of the molecules in container **X** is c_{rms} .

	X	Y
Volume of container / m³	3.0	2.0
Mass of gas / kg	0.50	1.50
Pressure in container / 10⁵ Pa	1.0	6.0

What is the rms speed of the molecules in container **Y**?

[1 mark]

A $0.50c_{\text{rms}}$

☐

B $1.15c_{\text{rms}}$

☐

C $1.41c_{\text{rms}}$

☐

D $2.00c_{\text{rms}}$

☐

Don't write

The radius of a lead-208 nucleus is 6.7×10^{-15} m.

What is the radius of an oxygen-16 nucleus?

[1 mark]

A 3.0×10^{-18} m

☐

B 5.1×10^{-16} m

☐

C 1.9×10^{-15} m

☐

D 2.8×10^{-15} m

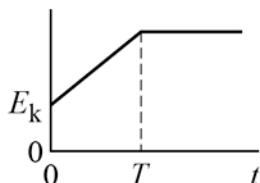
☐

A liquid is heated **at its boiling point** and at time T the liquid has completely boiled away. The gas continues to be heated.

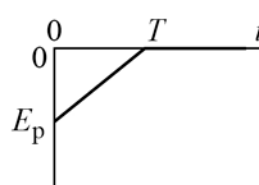
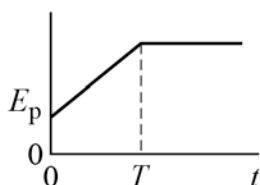
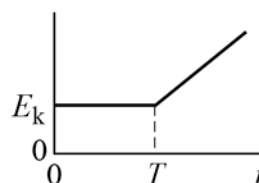
Which pair of graphs shows the variation with time t of the mean molecular kinetic energy E_k and the mean molecular potential energy E_p ?

[1 mark]

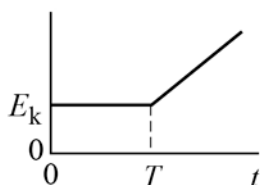
A



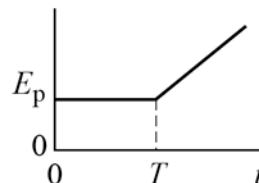
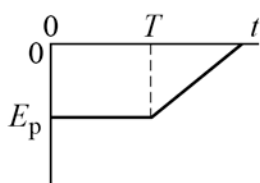
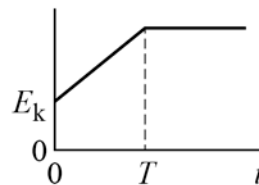
B



C



D



A

☐

B

☐

C

☐

D

☐

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1 5

Which row gives the measured variables and the controlled variable for Boyle's law or Charles's law?

[1 mark]

	Law	Measured variables	Controlled variable	
A	Boyle's law	pressure and temperature	volume	☐
B	Boyle's law	volume and temperature	pressure	☐
C	Charles's law	pressure and temperature	volume	☐
D	Charles's law	volume and temperature	pressure	☐



Don't write

1**6***Do not write
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A spherical flask is filled with an ideal gas and has a diameter of 1.66 m. The gas has a pressure of 1.43×10^5 Pa and a temperature of 15 °C.

What is the quantity of gas in the flask?

[1 mark]

A 1.43×10^2 mol

☐

B 1.15×10^3 mol

☐

C 2.75×10^3 mol

☐

D 2.20×10^5 mol

☐

What is the unit for $\frac{3R}{2N_A}$?

[1 mark]

A $\text{J mol}^{-1} \text{K}^{-1}$

☐

B J K^{-1}

☐

C $\text{J m}^{-3} \text{K}^{-1}$

☐

D $\text{J kg}^{-1} \text{K}^{-1}$

☐

Turn over for the next question

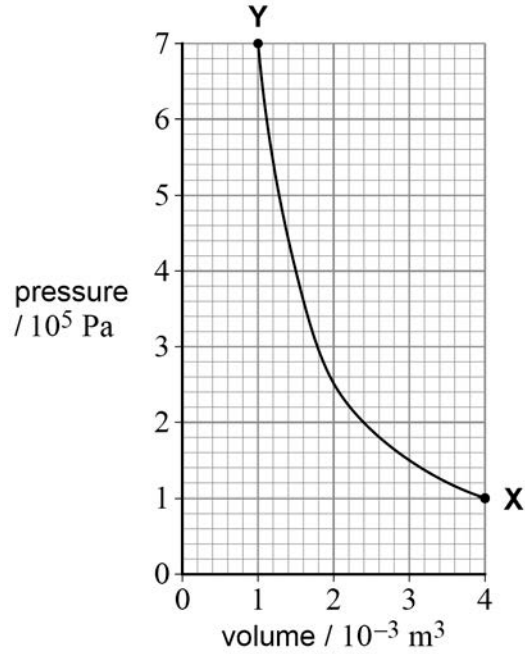
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1 8 The graph shows the variation of pressure with volume for a fixed mass of gas as it changes from condition **X** to condition **Y**.



Which row is correct for the change from **X** to **Y**?

[1 mark]

	Expansion or compression	Temperature of the system	
A	expansion	constant	☐
B	expansion	increase	☐
C	compression	constant	☐
D	compression	increase	☐

Which set of processes could be carried out at constant temperature on an ideal gas?

[1 mark]

A compression and heat transfer to the gas

☐

B compression and heat transfer from the gas

☐

C expansion and no heat transfer

☐

D expansion and heat transfer from the gas

☐

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box

A sample of gas contains oxygen and helium. Oxygen has a molar mass of 32 g and helium has a molar mass of 4 g.

What is $\frac{\text{rms speed of the oxygen molecules}}{\text{rms speed of the helium atoms}}$?

[1 mark]

A 0.12

☐

B 0.35

☐

C 2.8

☐

D 8.0

☐

0	1
---	---

The equation $pV = \frac{1}{3}Nm(c_{\text{rms}})^2$ applies to an ideal gas.

0	1	.	1
---	---	---	---

The derivation of the equation depends on assumptions made about the gas.

State **two** of those assumptions.

[2 marks]

1 _____

2 _____

0	1	.	2
---	---	---	---

A sample of an ideal gas has a temperature of 12.5 °C.

The mass of each particle of this gas is 6.6×10^{-26} kg.

Calculate c_{rms} (root mean square speed) for the particles in the gas.
Give your answer to an appropriate number of significant figures.

[4 marks]

$c_{\text{rms}} =$ _____ m s⁻¹



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0	1
---	---

Helium gas particles are single atoms that behave as an ideal gas.
A container of helium gas is used to inflate an empty balloon.

Immediately after inflation, the gas in the balloon has:

- a pressure of 106 kPa
- a temperature of 288 K
- a volume of $1.40 \times 10^{-2} \text{ m}^3$.

0	1	.	1
---	---	---	---

Show that the number of atoms in the balloon is approximately 3.7×10^{23}

[2 marks]

0	1	.	2
---	---	---	---

Each helium atom has a mass of $6.6 \times 10^{-27} \text{ kg}$.

Calculate the density of the gas in the balloon.

[2 marks]

density = _____ kg m^{-3}



0 1 . 3

The gas in the container has an initial temperature greater than 288 K.

Explain, with reference to the first law of thermodynamics, why the temperature of the gas decreases as the balloon is inflated.

[4 marks]

8

Turn over for the next question

Turn over ►



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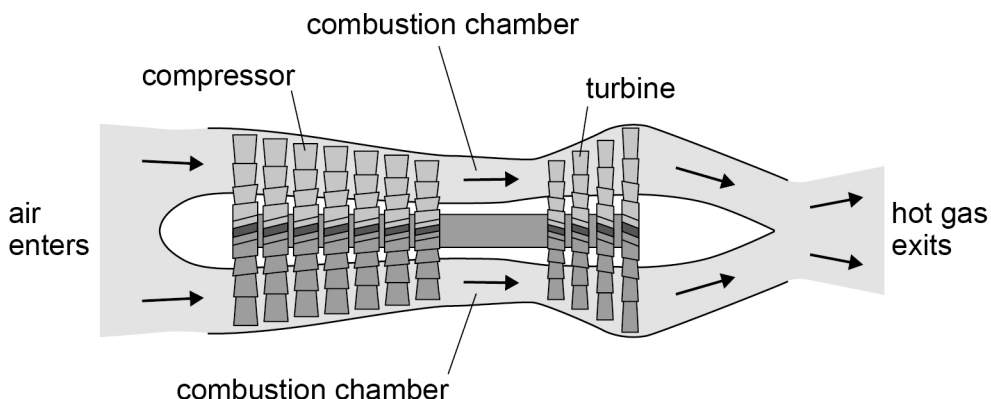
0 6

Figure 7 shows the flow of gases through a jet engine.

A jet engine has three sections:

- a compressor
- a combustion chamber
- a turbine.

Figure 7



Air enters the compressor at a pressure p_1 and an absolute temperature T_1 .

The compressor rapidly compresses the air to a pressure p_2 and an absolute temperature T_2 .

For this compression, $p_2 = 40p_1$.

Assume that the air behaves as an ideal gas.

0 6

. 1

The air enters the compressor with a density d_1 and exits the compressor with an increased density d_2 .

Explain, with reference to the ideal gas equation, why $\frac{T_2}{T_1} < 40$ for this increase in density.

[2 marks]



0 6 . 2

Explain, with reference to the first law of thermodynamics, why the temperature of the air increases as a result of this rapid compression.

[3 marks]

0 6 . 3

The pressures and temperatures of the air before and after this rapid compression are related by:

$$\frac{p_1}{p_2} = \left(\frac{T_1}{T_2} \right)^{3.5}$$

Calculate T_2 when $T_1 = 300 \text{ K}$.

[3 marks]

$T_2 =$ _____ K

Question 6 continues on the next page

Turn over ►

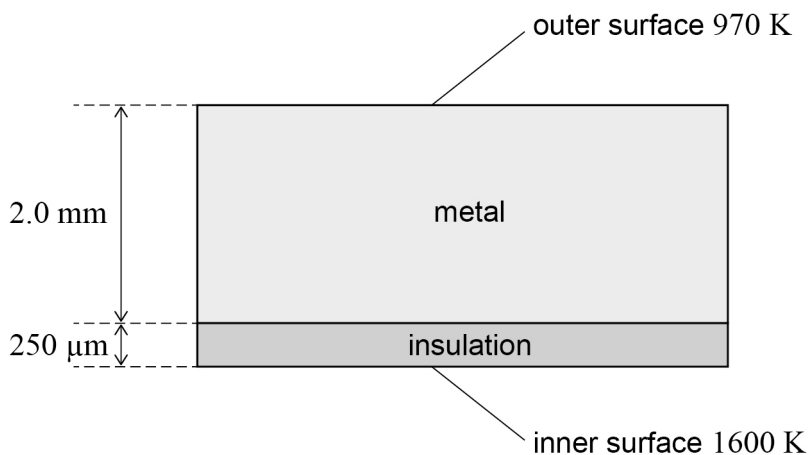


In the combustion chamber, fuel is added to the compressed air. The air–fuel mixture burns to increase the gas temperature even more.

0 6 . 4

Figure 8 shows a cross-sectional view of part of the wall of the combustion chamber. The wall of the combustion chamber is made of a metal and a layer of insulation. The inner surface of the wall is at a temperature of 1600 K. The outer surface is at a temperature of 970 K. The metal has a thickness of 2.0 mm and the insulation has a thickness of 250 μm .

Figure 8



The rate of energy transfer through the metal is the same as the rate of energy transfer through the insulation.

Determine the temperature at the boundary between the metal and the insulation.

thermal conductivity of metal = $27 \text{ W m}^{-1} \text{ K}^{-1}$

thermal conductivity of insulation = $0.74 \text{ W m}^{-1} \text{ K}^{-1}$

[3 marks]

temperature = _____ K



0 6 . 5

The gas from the combustion chamber drives a gas turbine that is rotating at 1.0×10^4 revolutions per minute. The turbine extracts energy from the gas at a rate of 19 MW.

Calculate the torque exerted by the gas on the turbine.

[3 marks]

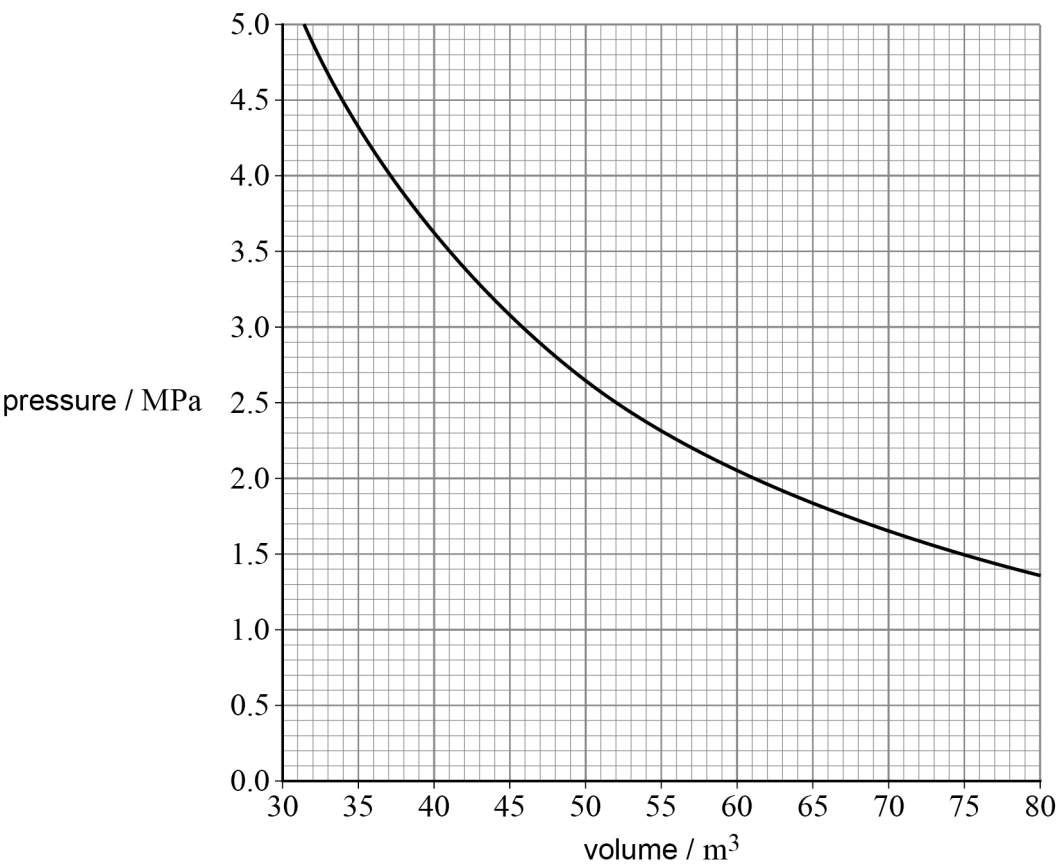
torque = _____ N m

Question 6 continues on the next page



A fixed mass of gas moves through the turbine.
Figure 9 shows the variation of pressure with volume for this gas.

Figure 9



The gas enters the turbine at 4.3 MPa and leaves at 1.5 MPa.
 Determine the work done by the fixed mass of gas on the turbine.

[3 marks]

work done by the gas = _____ J

END OF SECTION A



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.

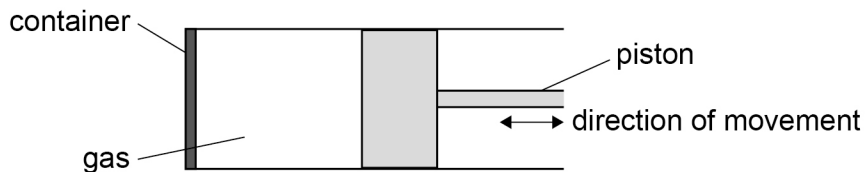


If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

A container holds a fixed mass of an ideal gas at a constant pressure. The container has a piston that is free to move horizontally.



Which values of ΔU and Q lead to the smallest change in the volume of the gas for this constant pressure?

[1 mark]

	$\Delta U / \text{J}$	Q / J	
A	-45	-60	☐
B	-15	+60	☐
C	+30	-45	☐
D	+60	+15	☐



Do not write Do not write

Which of these arises from a theoretical derivation rather than from an experimental observation?

[1 mark]

A Boyle's law

☐

B Charles's law

☐

C Brownian motion

☐

D the ideal gas equation

☐

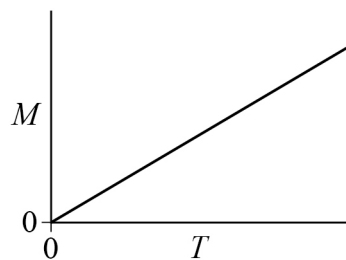
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The mass M of an ideal gas in a container is varied. The volume of the container is constant. For each mass M the temperature T of the gas is adjusted to keep the pressure constant.

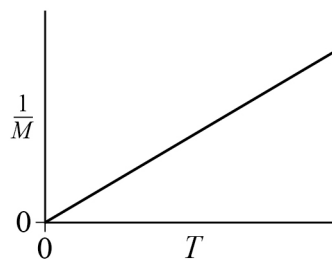
What is the variation of M with T ?

[1 mark]

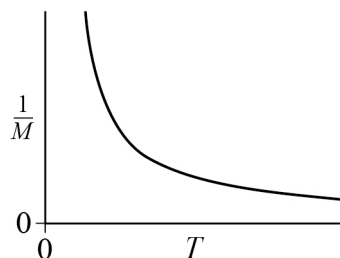
A



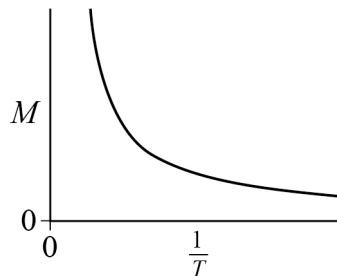
B



C



D



A

☐

B

☐

C

☐

D

☐

Turn over for the next question

Turn over ►



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1	1
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The molar mass of methane is 16 g mol^{-1} .

How many molecules are in 820 kg of methane?

[1 mark]

A 1.2×10^{25}

☐

B 3.1×10^{25}

☐

C 1.2×10^{28}

☐

D 3.1×10^{28}

☐

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1 **2** For one sample of an ideal gas, $c_{\text{rms}} = v$.

The temperature of the gas is changed at constant volume so that the pressure is halved.

What is the new value of c_{rms} ?

[1 mark]

A $\frac{v}{4}$ ☐

B $\frac{v}{2}$ ☐

C $\frac{v}{\sqrt{2}}$ ☐

D $\sqrt{2}v$ ☐



Don't write

Which pair of changes to the conditions of an ideal gas will always increase the internal energy of a fixed mass of the gas?

[1 mark]

	Gas pressure	Gas volume	
A	increase	increase	☐
B	increase	decrease	☐
C	decrease	increase	☐
D	decrease	decrease	☐

Turn over for the next question

Turn over ►



Do not write Do not write

Brownian motion provides evidence that gas is made of particles.

Brownian motion is demonstrated in a school laboratory by observing the motion of smoke particles under a microscope.

Properties of the gas molecules are deduced from observations made during the demonstration.

Explain **two** properties of the gas molecules in terms of these observations.

[4 marks]

1 _____

2 _____

Don't write

Gas bottles are used to store gases at high pressure.

Gas bottle **A** in **Figure 1** contains 25 mol of gas at a pressure of 1.0 MPa.

Assume that the interior of **A** is a cylinder of length 1.27 m and radius r , as shown in **Figure 2**, and that the gas behaves as an ideal gas.

Figure 1

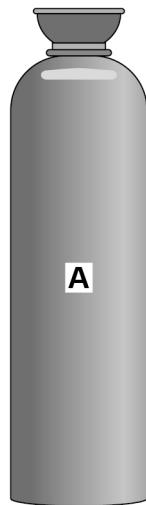
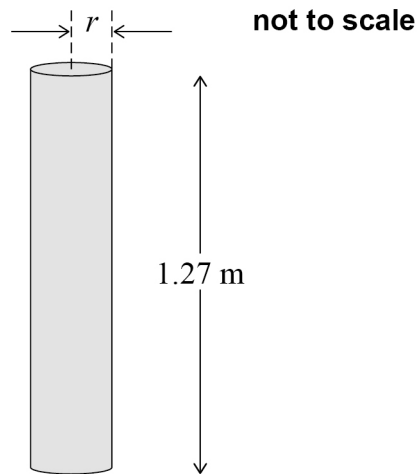


Figure 2



The temperature of the gas is 17 °C.

Determine r .

[3 marks]

$r =$ _____ m

Question 2 continues on the next page

Turn over ►



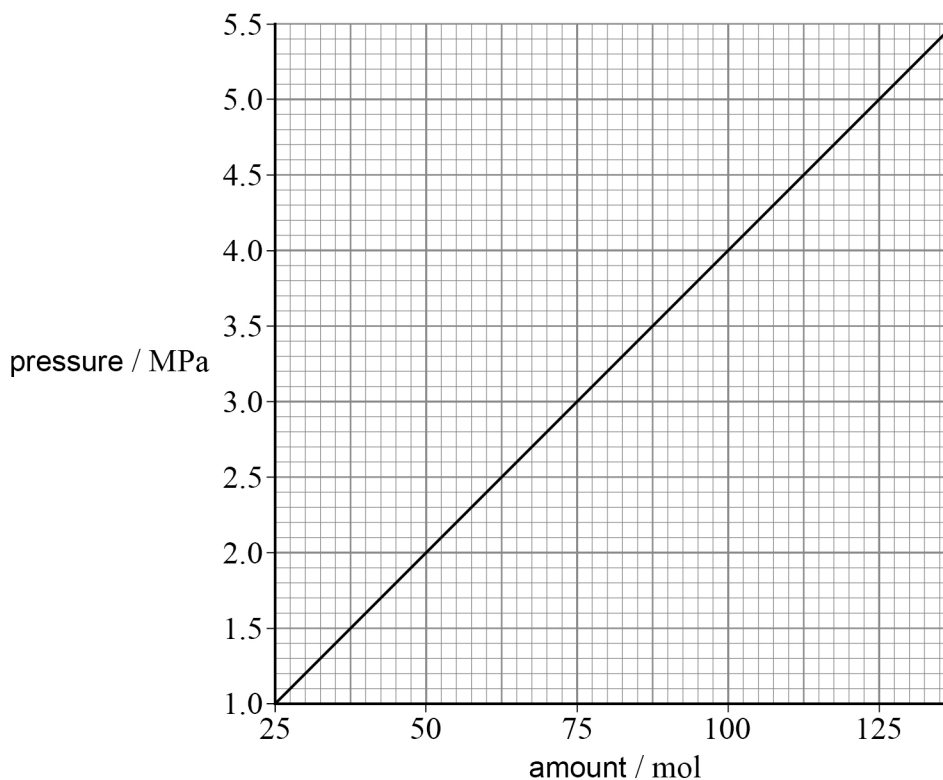
More of the gas is added to **A** and the pressure inside **A** increases.

The volume of **A** does not change.

The gas is put into **A** slowly so that the temperature of the gas remains constant at 17°C .

Figure 3 shows the variation of pressure with amount of gas in the bottle.

Figure 3



0 2 . 2

Calculate, using **Figure 3**, the mass of gas added to **A** to increase the pressure inside from 1.0 MPa to 3.8 MPa.

molar mass of the gas = 29 g mol^{-1}

[2 marks]

mass = _____ kg



0 2 . 3 Explain how **Figure 3** shows that the temperature of the gas is constant.

[3 marks]

0 2 . 4 Discuss the difference in the total internal energy of the ideal gas in **A** before and after the extra gas is added.

[4 marks]

0 2 . 5 Gas bottle **B** is identical to **A**.

B initially contains the same amount of gas as the initial amount in **A**.

The temperature of the gas in **B** is greater than 17 °C.

More of the gas is added to bottle **B** without changing its temperature.

Draw a line on **Figure 3** to show how the pressure in **B** changes as more gas is added.

[2 marks]



Do not write Do not write

The ideal gas equation is

$$pV = NkT$$

Which row identifies N and k ?

[1 mark]

	N	k	
A	the number of molecules in the gas	$\frac{R}{N_A}$	☐
B	the number of molecules in the gas	$\frac{N_A}{R}$	☐
C	the amount of gas, in mol	$\frac{N_A}{R}$	☐
D	the amount of gas, in mol	$\frac{R}{N_A}$	☐



Do not write Do not write

1

1

Which is **not** true for gas molecules in the kinetic theory of gases?

[1 mark]

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box

A Collisions between the molecules and container walls are elastic.

☐

B They have negligible mass.

☐

C There are no intermolecular forces except during collisions.

☐

D Their motion is random.

☐

03

A deuterium $\left(\begin{smallmatrix} 2 \\ 1 \end{smallmatrix} \text{H}\right)$ nucleus with an initial kinetic energy of $1.6 \times 10^{-14} \text{ J}$ approaches a second deuterium nucleus head-on. The second nucleus is initially stationary.

When the nuclei have reached a minimum separation d , each nucleus has a kinetic energy of $4.0 \times 10^{-15} \text{ J}$.

03.1

Calculate d .

[3 marks]

$d =$ _____ m

03.2

On another occasion, two deuterium $\left(\begin{smallmatrix} 2 \\ 1 \end{smallmatrix} \text{H}\right)$ nuclei have initial kinetic energies of $1.6 \times 10^{-14} \text{ J}$ each.

On this occasion, their minimum separation is different from the value of d in Question **03.1**.

Suggest, without calculation, how this minimum separation differs from the value of d in Question **03.1**.

[3 marks]



03.3

The minimum distances in Question **03.1** and Question **03.2** are both too large for the two nuclei to fuse.

Deuterium nuclei can fuse in a plasma in which the mean kinetic energy of nuclei is $1.6 \times 10^{-14} \text{ J}$.

Suggest why some deuterium nuclei can fuse in this plasma.

[1 mark]

03.4

Explain **two** reasons why the kinetic theory of gases can model the behaviour of an ideal gas but cannot model the behaviour of a plasma.

[2 marks]

1

2

03.5

An ideal gas has particles with a mean kinetic energy of $1.6 \times 10^{-14} \text{ J}$.

Calculate the absolute temperature of the gas.

[2 marks]

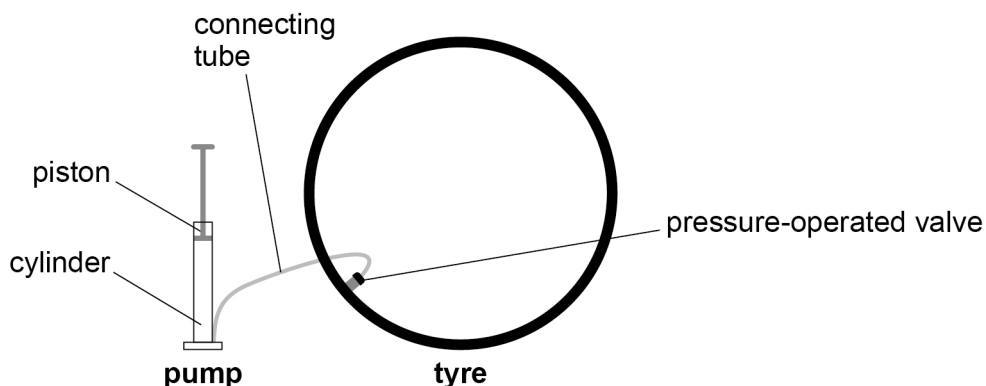
absolute temperature = _____ K



Don't write

Figure 4 shows a pump connected to a bicycle tyre through a connecting tube and a pressure-operated valve.

Figure 4

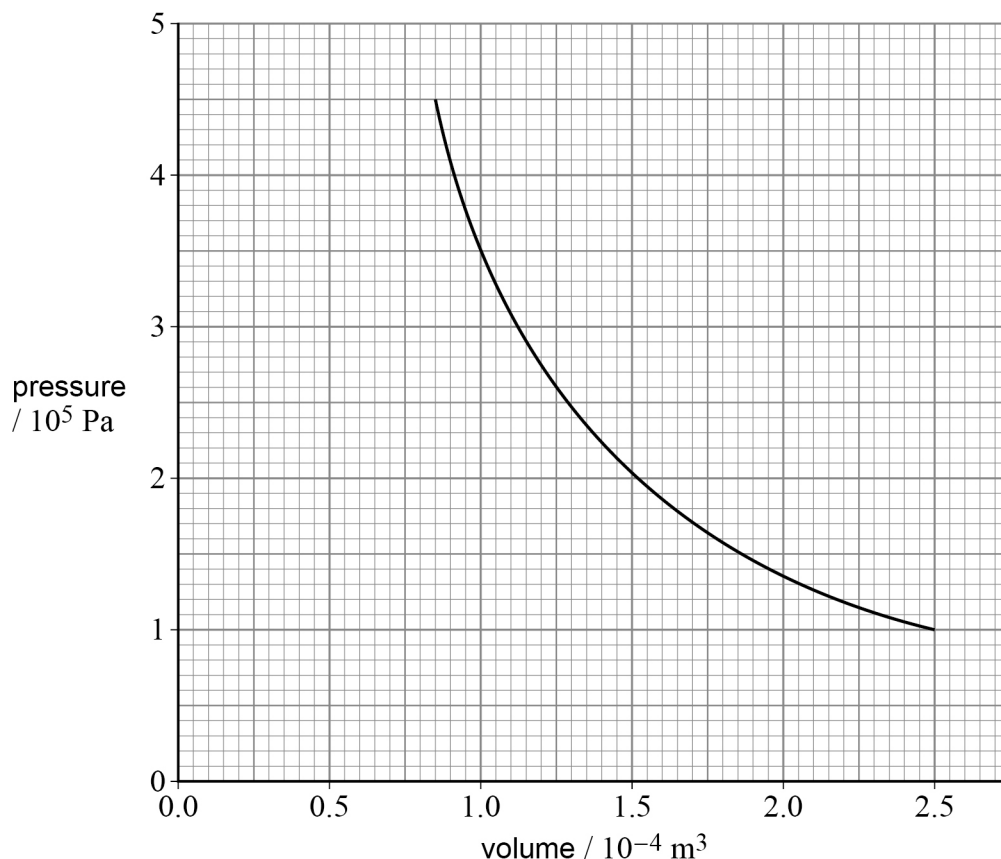


Initially, the piston is at the top of the cylinder as shown in **Figure 4**. The cylinder contains $2.5 \times 10^{-4} \text{ m}^3$ of air at a pressure of $1.0 \times 10^5 \text{ Pa}$ and a temperature of 17°C . The initial pressure of the air in the tyre is $4.5 \times 10^5 \text{ Pa}$.

When the pump is operated for the first time, the piston is pushed down quickly. All the air remains in the cylinder until the cylinder pressure is $4.5 \times 10^5 \text{ Pa}$. The valve opens when the cylinder pressure and the tyre pressure are equal. All the air in the cylinder then enters the tyre.

Figure 5 shows the variation of pressure with volume for the air in the cylinder.

Figure 5



0 6 . 1

Determine the work done in compressing the air in the cylinder to a pressure of 4.5×10^5 Pa.

[3 marks]

work done = _____ J

0 6 . 2

Calculate, in K, the temperature of the air in the cylinder when the pressure has increased to 4.5×10^5 Pa.

[2 marks]

temperature = _____ K

Question 6 continues on the next page

Turn over ►



0 6 . 3

The tyre needs an additional 0.27 mol of air to be inflated to its correct pressure. The volumes of the connecting tube and the valve are negligible.

Calculate the number of times the pump has to be operated to inflate the tyre.

[3 marks]

number = _____

0 6 . 4

Explain, with reference to the first law of thermodynamics, why the temperature of the air in the cylinder increases when the piston is pushed down quickly.

[3 marks]

END OF SECTION A



Turn over for Section B

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

A non-ideal gas has molecules that contain more than one atom.
Each molecule has a moment of inertia.

The internal energy of the gas is the sum of:

[1 mark]

- A** the translational kinetic energies of all of the molecules.
- B** the translational, rotational and vibrational kinetic energies of all of the molecules.
- C** the potential energies and the translational kinetic energies of all of the molecules.
- D** the potential energies and the translational, rotational and vibrational kinetic energies of all of the molecules.

☐☐☐☐

Do not write

1 1

Atmospheric pressure is 100 kPa.

Sample **X** of an ideal gas has a volume of 2.0 m^3 and an initial pressure of 100 kPa above atmospheric pressure. It is compressed at constant temperature until its volume is V_X and its pressure is 300 kPa above atmospheric pressure.

Sample **Y** of an ideal gas has a volume of 2.0 m^3 and an initial temperature of 177°C . It is heated at constant pressure until its temperature is 354°C and its volume is V_Y .

What are V_X and V_Y ?

[1 mark]

	V_X / m^3	V_Y / m^3	
A	1.0	2.8	☐
B	1.0	4.0	☐
C	0.67	2.8	☐
D	0.67	4.0	☐



Don't write

1	2
---	---

A student uses a microscope to observe Brownian motion in some smoke.

Which statement is **not** true?

[1 mark]

A Randomly moving air molecules are in collision with smoke particles.



B Smoke particles are seen to move randomly.



C Air molecules are seen to move randomly.



D The random motion of smoke particles demonstrates the random motion of air molecules.

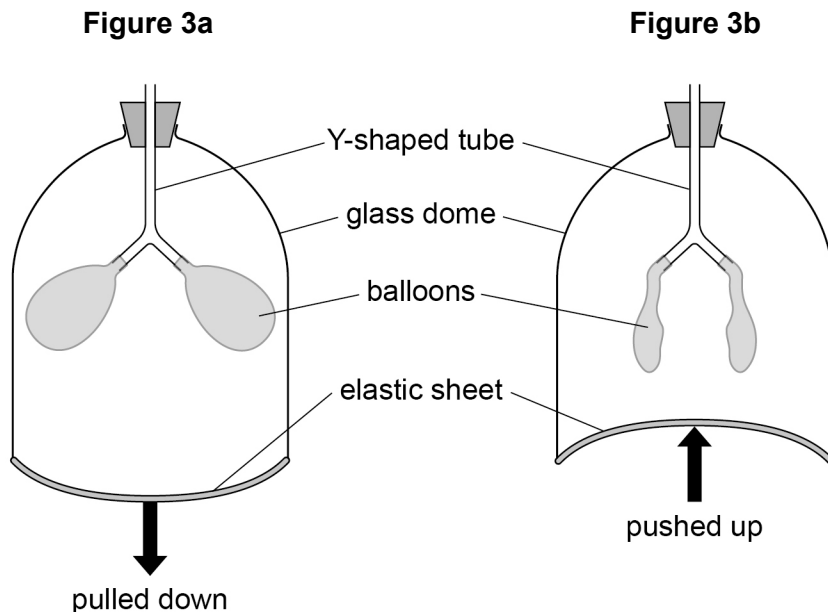


Figures 3a and 3b show a simple model of the human breathing system.

The model consists of a glass dome that contains two balloons.

The balloons are connected by a Y-shaped tube that is open to the atmosphere at one end.

Air can be made to enter or leave the balloons by moving an elastic sheet that covers the bottom of the dome.



The air is heated as it travels through the Y-shaped tube. The heater is not shown. The air that enters the balloons is at a temperature of 32°C .

02.1 The temperature of the air before it enters the Y-shaped tube is 15°C .

Calculate the change in the average molecular kinetic energy of air when its temperature is increased from 15°C to 32°C .

[2 marks]

change in average molecular kinetic energy = _____ J

Question 2 continues on the next page

Turn over ►



The elastic sheet is pulled down as shown in **Figure 3a** and then pushed up as shown in **Figure 3b**.

Air moves in and out of the balloons so that the air inside the balloons is always at atmospheric pressure.

Table 2 shows data for the air in one balloon.

Table 2

	Figure 3a	Figure 3b
volume / 10^2 cm^3	3.1	0.35
temperature / $^{\circ}\text{C}$	32	32

Atmospheric pressure is $1.10 \times 10^5 \text{ Pa}$.
Assume that the air behaves as an ideal gas.

0 2 . 2

Calculate the work done on the air as it leaves **one** balloon.

[2 marks]

work done = _____ J



0 2 . 3

Calculate the number of particles of air that leave **one** balloon when the elastic sheet is pushed up.

[3 marks]

number of particles = _____

0 2 . 4

The model is taken to the top of a mountain.

The air in the balloons is now at a much smaller atmospheric pressure.

The air entering the Y-shaped tube is also at a much lower temperature.

The air is now heated to a temperature of 32 °C as it travels through the tube, as in Question **02.1**.

Discuss, without calculation, how taking the model to the top of the mountain affects the energy required to:

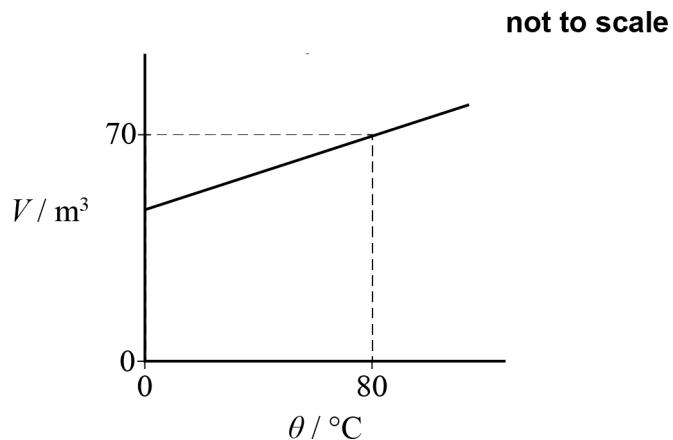
- move the same volume of air as in Question **02.2**
- heat the same number of air particles to the same temperature as in Question **02.3**
- move the same number of particles out of **one** balloon as calculated in Question **02.3**.

[3 marks]



Do not write

The graph shows the variation of volume V with temperature θ for a fixed mass of an ideal gas at constant pressure.



What is the gradient of the line?

[1 mark]

A 0.20

☐

B 0.26

☐

C 0.58

☐

D 0.88

☐

Do not write

An equation that arises from the kinetic theory model is

$$pV = \frac{1}{3}Nm(c_{\text{rms}})^2$$

Which assumption is required in the derivation of this equation?

[1 mark]

A The particles all move with the same speed.

☐

B The mass of an individual particle is negligible.

☐

C The sides of the container all have the same area.

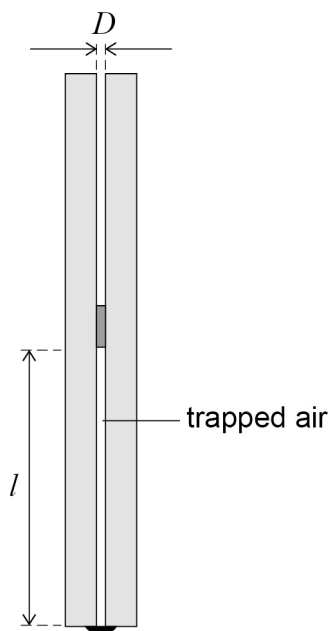
☐

D The volume of an individual particle is negligible.

☐

1 3

A student does a Charles's law experiment. The student uses measurements of the length of air trapped in a tube to estimate a value for absolute zero.



The initial length of the trapped air is l and the diameter of the tube is D .

The behaviour of the liquid trapping the air does not depend on the diameter of the tube.

Which row shows the changes to l and D that could lead to an improvement in the accuracy of the student's value?

[1 mark]

	l	D	
A	decrease	increase	☐
B	increase	decrease	☐
C	decrease	decrease	☐
D	increase	increase	☐



Do not write

The energy storage capacity of a pumped storage station is 9.1×10^6 kW h.
At peak output, the rate of flow of water through the generators is $390 \text{ m}^3 \text{ s}^{-1}$.
The generators are 110 m below the upper reservoir.
The density of water is 1000 kg m^{-3} .

What is the maximum possible length of time for which the station can provide peak output?

[1 mark]

A $1.3 \times 10^3 \text{ s}$

☐

B $2.2 \times 10^4 \text{ s}$

☐

C $7.8 \times 10^4 \text{ s}$

☐

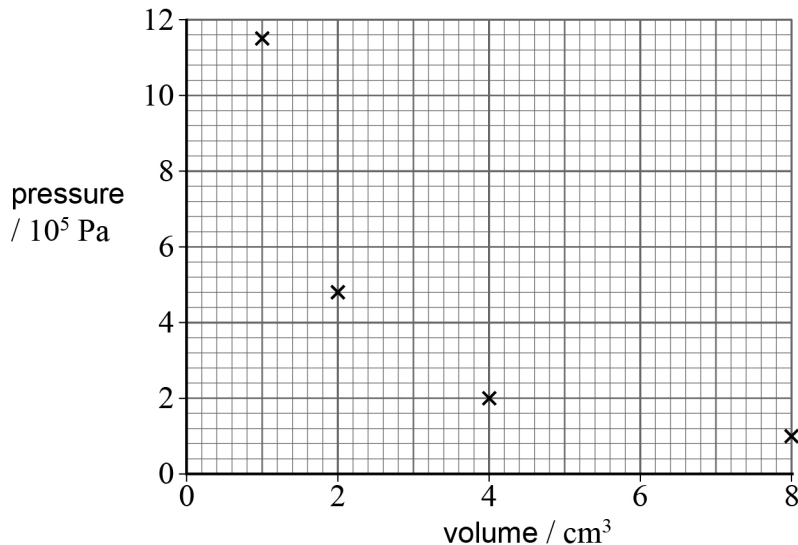
D $7.6 \times 10^5 \text{ s}$

☐

Questions **18** and **19** relate to the data shown below.

The pressure on a fixed mass of ideal gas is increased.

The graph shows the values of pressure and volume for the gas.



The temperature of the gas is 35°C when its volume is 8 cm^3 .

1**8**

What is the number of moles of molecules in the gas?

[1 mark]

A 3.1×10^{-4}

☐

B 2.8×10^{-3}

☐

C 3.1×10^{-3}

☐

D 2.8×10^{-2}

☐

Which volume is the smallest at which the temperature of the gas remains at 35 °C?

[1 mark]

A 8 cm³

☐

B 4 cm³

☐

C 2 cm³

☐

D 1 cm³

☐

Turn over for the next question

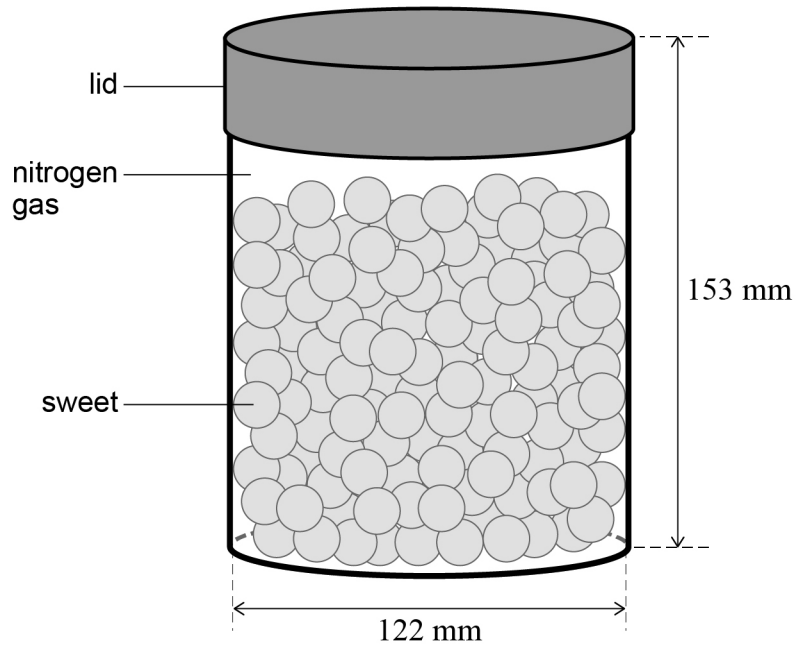
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Do not write

Figure 1 shows a container in the shape of a cylinder. It contains identical spherical sweets and nitrogen gas. The lid seals the container so that gas cannot move in or out.

Figure 1



There is $1.15 \times 10^{-3} \text{ m}^3$ of nitrogen in the container.

The container has an internal height of 153 mm and the base has an internal diameter of 122 mm.

There are 595 sweets in the container.

The density of the sweets is $1.18 \times 10^3 \text{ kg m}^{-3}$.



0 1 . 1 Calculate the mass of one sweet.

[3 marks]

mass of one sweet = _____ kg

The nitrogen in the container is at a temperature of 291 K and a pressure of 1.1×10^5 Pa.

0 1 . 2 Calculate, in g, the mass of nitrogen in the container.

mass of 1.0 mol of nitrogen = 28 g

[2 marks]

mass = _____ g

Question 1 continues on the next page

Turn over ►



Do not write